

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. We separate the weighted strict topology, whose dual is the vector space of signed Radon measures, from the stronger Hölder/graph topology, whose dual may contain distributions. The invariant-integral nullspace excludes constants, and full-pressure rigidity is proved by platform-specific Sinai Livšic or finite-horizon balance-coboundary arguments. A uniformly equivalent real coercive form norm yields one common type-(B) domain for generator and compressed-memory response. Resolved projections are treated separately, source and state derivatives have distinct chain rules, and conditional-kernel convergence gives the optional-projection, Girsanov, and quadratic-BSDE representation.

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CONTENTS

1. Round-fourteen positive closure: two-topology cotangents and a common form bundle	2
1.1. The weighted strict quotient	2
1.2. The strong response space	2
1.3. Full pressure-functional rigidity	3
1.4. A real common closed form domain	3
1.5. Resolved observables and memory response	4
1.6. Typed contraction derivatives	5
1.7. Optional projections and stochastic representations	5
Scope and verification status	6

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1. ROUND-FOURTEEN POSITIVE CLOSURE: TWO-TOPOLOGY COTANGENTS AND A COMMON FORM BUNDLE

The measure-theoretic cotangent topology and the strong spectral/graph topology are not identified. The first has invariant signed Radon measures as its dual; the second has distributional duals and is used only for response. Memory is differentiated through a uniformly coercive family of closed forms on one real common domain.

1.1. The weighted strict quotient. Let E be a Polish path space and $W \geq 1$ a continuous coercive weight. Put

$$C_W(E) = \{F \in C(E) : F/W \text{ vanishes at weighted infinity}\},$$

with the weighted strict topology β_W . Let $\mathcal{M}_W(E)$ be the signed Radon measures with $\int W d|\mu| < \infty$. Define

$$\mathcal{N}_{\text{inv}} = \overline{\text{span}}^{\beta_W} \{U - U \circ \Theta_t : U \in C_W(E), t \in \mathbb{Q}_+\}.$$

Constants are not included.

Theorem 1.1 (Invariant signed-Radon dual). *The continuous dual of $C_W(E)$ is $\mathcal{M}_W(E)$. The continuous dual of $C_W(E)/\mathcal{N}_{\text{inv}}$ is the vector space*

$$\mathcal{M}_W^{\text{inv}}(E) = \{\mu \in \mathcal{M}_W(E) : \Theta_{t\#}\mu = \mu, t \geq 0\}.$$

Invariant probabilities form only the normalized positive slice of this vector space. Two functions have the same integral under every invariant probability exactly when their difference lies in $\mathcal{N}_{\text{inv}}$; equality up to one common constant corresponds to $\mathbb{R}1 + \mathcal{N}_{\text{inv}}$.

Proof. The weighted Riesz representation theorem identifies every β_W -continuous functional with a signed Radon measure of finite W -moment. The annihilator of the closed coboundary span consists exactly of measures satisfying $\mu(U) = \mu(U \circ \Theta_t)$ for rational t , and continuity in time extends this to all t . Hahn–Banach gives the converse annihilator identity. Testing the constant function distinguishes equality of all integrals from equality modulo one common constant. $\square$

1.2. The strong response space. For each platform let $\mathcal{A}_W^m \subset C_W(E)$ be its weighted Hölder/anisotropic/graph algebra. Its norm contains dynamical regularity, so its continuous dual may contain distributions. Let

$$\iota_m : \mathcal{A}_W^m \longrightarrow C_W(E)$$

be the continuous inclusion.

Proposition 1.2 (Two-topology cotangent functor). *Pullback by ι_m embeds invariant Radon cotangents into $(\mathcal{A}_W^m)^*$, but does not exhaust the strong dual. Variational phases are taken in the strict Radon dual, while spectral derivatives and seam/contact currents are taken in the strong dual. Every crossing between the two is the explicit pullback ι_m^* .*

Proof. Continuity of ι_m gives the dual pullback. Derivatives of point evaluation, seam traces, and graph boundary forms are continuous on the strong space but need not be Radon measures, showing that the inclusion can be strict. The stated typing is therefore forced. $\square$

1.3. Full pressure-functional rigidity. For a potential F , let

$$Q_F(H) = P(F + H) - P(F)$$

on a strong source neighborhood.

Theorem 1.3 (Platform-typed rigidity). *Suppose $Q_F(H) = Q_G(H)$ for every H in a neighborhood of zero. Then the equilibrium linear functionals of F and G agree on the strict Radon quotient.*

On the mixing Sinai symbolic platform, if F, G are dynamically Hölder, then

$$F - G = c + U - U \circ \Theta$$

by the periodic-orbit Livšic theorem. On the finite-horizon hard-sphere platform, the corresponding conclusion is the full balance-coboundary identity of B3 on the declared short-time source space. No platform-wide Livšic theorem is imported from one model to the other.

Proof. Differentiate the identity at every small H . The exposed equilibrium functional is the derivative of pressure, so the two families of equilibrium Radon functionals agree. At $H = 0$ their invariant integrals agree on all directions.

For the Sinai coding, equality of the Gibbs functional under all local perturbations implies equality of conditional Jacobians. Taking products around a periodic word shows that the periodic sums of $F - G$ equal one constant times the period. The Hölder Livšic construction on the mixing Markov extension gives the coboundary.

For hard spheres, B3 proves that a source annihilating every local covariance and every balanced variation is an endpoint/transport/collision coboundary. Applying that theorem to the derivative identity gives the stated model-specific quotient. $\square$

Equality of one scalar pressure value is never used.

1.4. A real common closed form domain. Fix a regular parameter chart $\eta \in U$. On an algebraic cylinder core $\mathcal{C}$, let a_η be the sectorial form of the Doob or kinetic generator. Assume the estimates proved in A4/B4:

$$\begin{aligned} \Re a_\eta(F, F) + c\|F\|_\eta^2 &\asymp \Re a_{\eta_0}(F, F) + c\|F\|_{\eta_0}^2, \\ |a_\eta(F, G) - a_{\eta_0}(F, G)| &\leq \epsilon\|F\|_V\|G\|_V, \end{aligned}$$

uniformly on the chart. Define the real norm

$$\|F\|_V^2 = \Re a_{\eta_0}(F, F) + c\|F\|_{\eta_0}^2$$

and let V be the completion of $\mathcal{C}$.

Theorem 1.4 (Common-domain type-(B) form bundle). *The embedding $V \hookrightarrow H_\eta$ is dense and continuous for every $\eta \in U$, and the closures of a_η all have the same form domain V . The maps*

$$\eta \mapsto a_\eta \in \mathcal{L}(V, V^*)$$

are C^q , and

$$D(z - L_\eta)^{-1} = (z - L_\eta)^{-1}(DL_\eta)(z - L_\eta)^{-1}$$

holds as a bounded composition $V^ \rightarrow V \rightarrow V^* \rightarrow V$. The norm uses the real coercive part, never the generally complex number $a_\eta(F, F)$.*

Proof. Uniform equivalence of the coercive real form norms makes the completion of $\mathcal{C}$ independent of η . Since $\mathcal{C}$ is the graph core proved in A4 or B4, it is dense in each H_η ; equivalence preserves density. The relative form estimates and their parameter derivatives give bounded maps $V \rightarrow V^*$. Kato's type-(B) form theorem constructs the m-sectorial generators and their resolvents. Differentiate the weak resolvent equation

$$z(u, v)_{H_\eta} + a_\eta(u, v) = \langle f, v \rangle$$

on the fixed domain V to obtain the formula. $\square$

This avoids both an impossible regularity-losing Kato ODE and an unsupported finite-Gram identification of the full generator domain.

1.5. Resolved observables and memory response. Let $R_\eta \subset V$ be a finite-dimensional family of physical resolved observables after quotienting constants and conserved null modes. Its correlation Gram matrix is uniformly positive on a smaller chart, so a C^q orthonormal frame exists. Let P_η be the corresponding finite-rank projection in H_η .

Theorem 1.5 (Covariant compressed-memory derivative). *With the common form-domain resolvent of Theorem 1.4, define*

$$C_\eta(z) = P_\eta(z - L_\eta)^{-1}P_\eta, \quad \widehat{K}_\eta(z) = zP_\eta - P_\eta L_\eta P_\eta - C_\eta(z)^{-1}.$$

On every compact subset avoiding the genuine Schur-complement poles, $\widehat{K}_\eta(z)$ is C^{q-1} in η . Its derivative contains the three explicit terms

$$DP_\eta, \quad (z - L_\eta)^{-1}(DL_\eta)(z - L_\eta)^{-1}, \quad -C_\eta^{-1}(DC_\eta)C_\eta^{-1}.$$

After the A4 principal-part decomposition, differentiation commutes with inverse Laplace transformation in the exponentially weighted distribution space.

Proof. The finite Gram matrix gives a smooth projection because its smallest eigenvalue is uniformly positive after the null quotient. Apply the form resolvent identity and the ordinary inverse-matrix rule. A4 supplies integrable vertical bounds for the final Schur complement after pole-zero cancellation; the same bounds hold for one parameter derivative by the form estimate. Dominated Bromwich inversion proves the last assertion. $\square$

1.6. Typed contraction derivatives. Let $\mathcal{C} : X \rightarrow Y$ be a measurable state contraction and let

$$Q_{\mathcal{C}}(\varphi) = Q(\varphi \circ \mathcal{C}).$$

Theorem 1.6 (Source and state chain rules). *For a source perturbation $h : Y \rightarrow \mathbb{R}$,*

$$DQ_{\mathcal{C}}(\varphi)[h] = DQ(\varphi \circ \mathcal{C})[h \circ \mathcal{C}].$$

If separately $\mathcal{C}_{\eta}$ and φ_{η} vary with a parameter, then the state derivative is

$$\frac{d}{d\eta}(\varphi_{\eta} \circ \mathcal{C}_{\eta}) = (\partial_{\eta}\varphi_{\eta}) \circ \mathcal{C}_{\eta} + D\varphi_{\eta}(\mathcal{C}_{\eta}) \partial_{\eta}\mathcal{C}_{\eta}.$$

The two formulas are not identified. Their second derivatives are obtained by differentiating the same typed maps.

Proof. The first identity is the chain rule for the linear pullback of source functions. The second is the ordinary state/parameter chain rule. Their domains are different, which proves the typing assertion. $\square$

1.7. Optional projections and stochastic representations. Let M_t^{ε} be the exact complete-history likelihood martingale and let $\mathcal{G}_t^{\varepsilon}$ be the resolved filtration.

Theorem 1.7 (Resolved likelihood limit). *Assume the A4 conditional-kernel rough convergence and the B3 uniform $L^{1+\delta}$ likelihood bound. Then*

$$\mathbb{E}[M_t^{\varepsilon} \mid \mathcal{G}_t^{\varepsilon}]$$

converges stably to the optional projection of the limiting Cameron–Martin exponential. On the nondegenerate resolved Brownian filtration this projection is the Girsanov stochastic exponential, and its logarithmic value solves the quadratic BSDE with the declared terminal condition.

Proof. A4 gives joint convergence of conditional kernels and enhanced paths; the uniform $L^{1+\delta}$ bound gives uniform integrability. Approximate the conditional expectation by bounded finite-history cylinders, pass to the limit, and remove the approximation using the kernel convergence. The Brownian martingale representation theorem identifies the optional projection as a stochastic exponential. Itô’s formula for its logarithm gives the quadratic BSDE, whose bounded solution is unique by exponential transformation. $\square$

Theorem 1.8 (Round-fourteen C2 closure). *The series has a correctly typed invariant signed-Radon quotient, a separate strong response dual, platform-specific full-pressure rigidity, a real common closed form bundle for memory response, and distinct source/state contraction derivatives with the resolved likelihood representation.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, 1.6, and 1.7, with Proposition 1.2. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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